

Nuclear Chemistry and Applications of Radioactivity

Nuclear Chemistry

Nuclear chemistry is the branch of chemistry that deals with **changes occurring in the nucleus of atoms** and the radiation produced during these changes.

Nuclear reaction / nuclear transformation

A **nuclear reaction** is a process in which the **nucleus of an atom changes**, often producing a different nucleus or even a different element.

Chemical reaction vs Nuclear reaction

Chemical reaction

Involves mainly **electrons**

Nucleus remains unchanged

Usually forms new compounds

Energy change is relatively small

Nuclear reaction

Involves the **nucleus**

Nucleus changes

Can form a **new element**

Energy released can be extremely large

Easy way to remember:

Chemical reaction → electrons change

Nuclear reaction → nucleus changes

Radioactivity

Radioactivity is the **spontaneous emission of radiation from an unstable atomic nucleus** in order to become more stable.

During radioactivity:

Unstable nucleus → emits radiation → more stable nucleus

Sometimes the new nucleus belongs to a **different element**. This change is called **nuclear transmutation**.

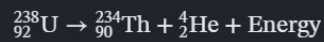
Why Uranium is Unstable

- **Large Size:** Uranium has 92 protons. All elements with 84 or more protons have nuclei that are too large for the strong nuclear force to keep them stable. [\[1\]](#), [\[2\]](#)
- **Electrical Repulsion:** Positive protons push away from each other. As the nucleus grows bigger, this push overcomes the tight binding of the nuclear force. [\[1\]](#)
- **Radioactive Decay:** To gain stability, unstable atoms like [Uranium](#) break apart or eject particles (such as alpha particles) to transform into lighter, stable elements like lead.

Important terms

- **Radioactive element:** An element having an unstable nucleus that spontaneously emits radiation.
- **Nuclear disintegration:** The breakdown/change of an unstable nucleus.
- **Radioactive radiation:** Radiation emitted from an unstable nucleus.
- Common radioactive elements: **Uranium (U), Thorium (Th), Radium (Ra), Polonium (Po)** etc.

The equation represents the **alpha decay** of Uranium-238, which is its most common form of natural radioactivity.



Reaction Components

- **Parent Nucleus (${}_{92}^{238}\text{U}$):** An unstable Uranium atom with 92 protons and 146 neutrons.
- **Daughter Nucleus (${}_{90}^{234}\text{Th}$):** A Thorium atom created after the decay, containing 90 protons and 144 neutrons.
- **Alpha Particle (${}_2^4\text{He}$):** A helium nucleus consisting of 2 protons and 2 neutrons shot out from the Uranium atom.
- **Energy:** Released as kinetic energy and gamma radiation during the breakdown.

Discovery of Radioactivity



Henri Becquerel

- Discovered **radioactivity in 1896**.
- He found that uranium salts emitted invisible radiation **even without exposure to sunlight**.
- He is regarded as the **father of radioactivity**.

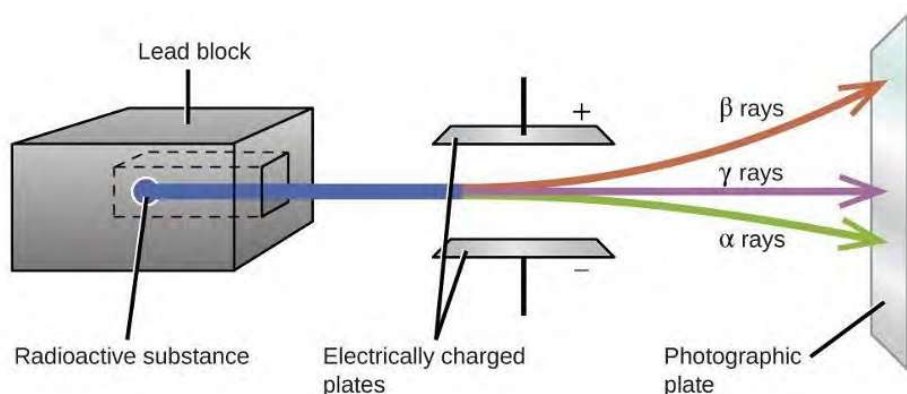
Marie Curie and Pierre Curie

- Further investigated radioactivity.
- Discovered the radioactive elements **polonium and radium**.
- **Henri Becquerel, Marie Curie and Pierre Curie** received the **1903 Nobel Prize in Physics**.
- Marie Curie later received the **1911 Nobel Prize in Chemistry** for her work on radium and polonium.

Types of Radioactive Radiation

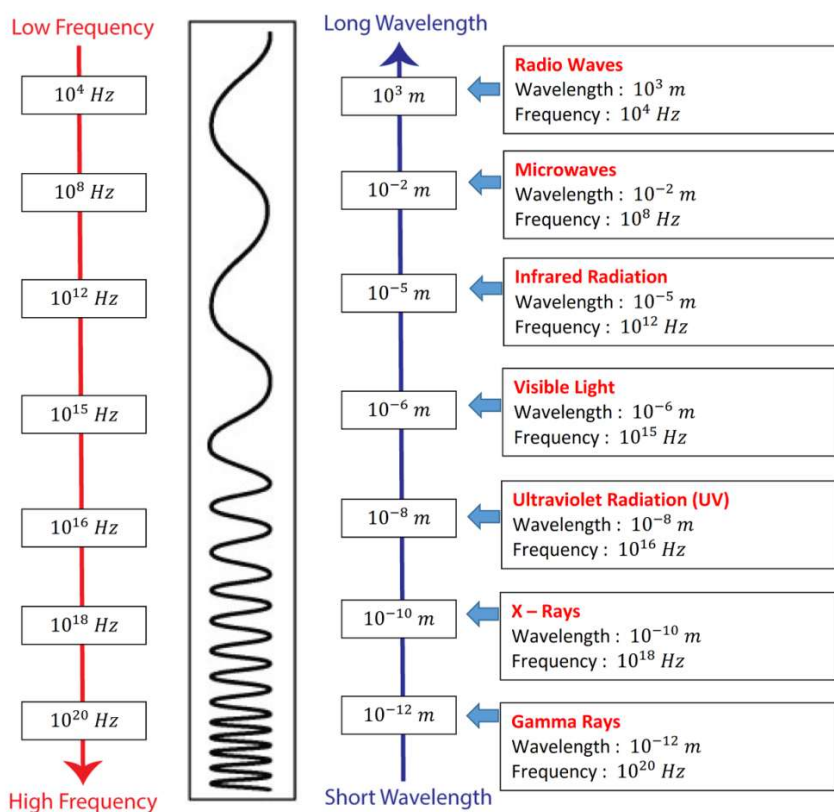
Radioactive substances mainly emit three types of radiation:

1. **Alpha (α)**
2. **Beta (β)**
3. **Gamma (γ)**



Comparison of α , β and γ rays

Property	α (Alpha)	β (Beta)	γ (Gamma)
Nature	Particle	Particle	Electromagnetic radiation
Charge	+2	-1	0
Mass	4 amu	Very small, about 1/1840 of H atom	0
Symbol	${}^4_2\text{He}^{2+} / \alpha$	${}^0_{-1}\text{e} / \beta$	γ
Ionizing power	Maximum	Medium	Very low
Penetrating power	Least	Medium	Very high
Velocity	About 1/10 of light	Nearly equal to light	Equal to light



Easy memory

Ionizing power:

$\alpha > \beta > \gamma$

Penetrating power:

$\gamma > \beta > \alpha$

So:

Alpha = strongest ionizer but weakest penetration

Gamma = weakest ionizer but strongest penetration

Why?

- **α particles** are relatively heavy and have a +2 charge, so they interact strongly with matter → **high ionization**, but lose energy quickly → **low penetration**.
- **β particles** are much lighter and have -1 charge → intermediate behavior.
- **γ rays** have no mass and no charge → interact less with matter → **low ionization but very high penetration**.

Units of Radioactivity

Radioactivity is measured by the **number of nuclear disintegrations occurring per unit time**.

1. Becquerel (Bq)

- SI unit of radioactivity.
- **1 disintegration per second = 1 Bq**

where **dps** = disintegrations per second.

2. Curie (Ci)

- A larger unit of radioactivity.
- **1 Ci = 3.7×10^{10} disintegrations per second**

3. Rutherford (Rd)

- Another unit of radioactivity.
- **1 million disintegrations per second = 1 Rutherford**

Quick comparison

Unit	Value
1 Bq	1 dps
1 Rd	10^6 dps
1 Ci	3.7×10^{10} dps

Writing Nuclear Equations

A **nuclear equation** represents the change occurring in the nucleus during a nuclear reaction or radioactive decay.

A nuclear species is written as:



where:

- **A = Mass number = number of protons + neutrons**
- **Z = Atomic number = number of protons**
- **X = Symbol of the element**

Important rule

In every nuclear equation:

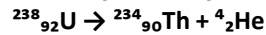
- **Total mass number is conserved.**
- **Total atomic number is conserved.**

Common Nuclear Particles

Particle	Symbol	Mass number (A)	Atomic number (Z)
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Alpha	${}^4_2\text{He}$ or α	4	2
Beta	${}^0_{-1}\text{e}$ or β	0	-1
Gamma	γ	0	0
Neutron	${}^1_0\text{n}$	1	0

Example: Alpha decay



Here:

- Mass number: $238 = 234 + 4$
- Atomic number: $92 = 90 + 2$

Therefore, uranium-238 changes into thorium-234 by emitting an **alpha particle**.

Natural and Artificial Radioactivity

Radioactivity is mainly of two types:

1. **Natural radioactivity**
2. **Artificial radioactivity**

1. Natural Radioactivity

Definition

Natural radioactivity is the spontaneous disintegration of an unstable nucleus occurring naturally, accompanied by the emission of radiation.

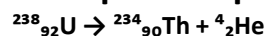
What does spontaneous mean?

It means the radioactive nucleus **decays by itself without any external trigger**.

Naturally radioactive elements/isotopes

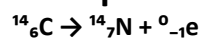
- Uranium-238
- Thorium-232
- Radium-226
- Polonium-210
- Carbon-14

Example 1: Alpha decay

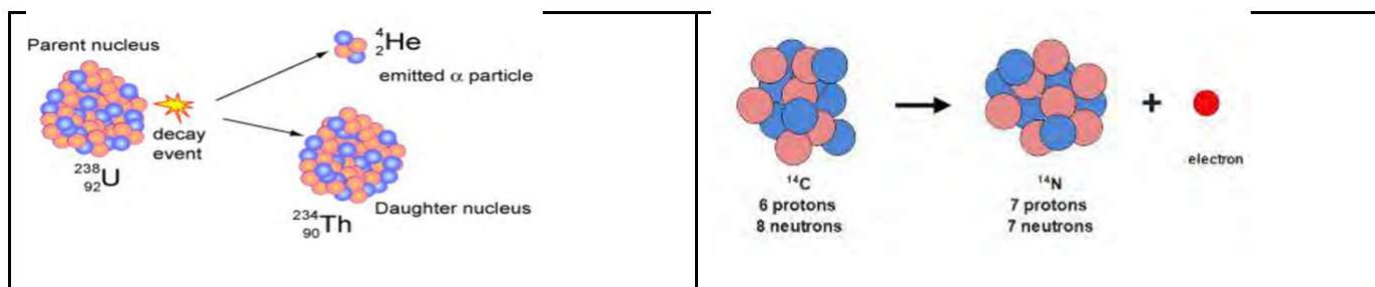


Uranium-238 naturally changes into thorium-234 by emitting an **alpha particle**.

Example 2: Beta decay



Carbon-14 changes into nitrogen-14 by emitting a **beta particle**.



Applications of Natural Radioactivity

- **Carbon dating:** Determining the age of fossils and archaeological materials.
- **Uranium dating:** Determining the age of rocks.
- **Medical imaging**
- **Smoke detectors**
- **Tracing movement of groundwater**

2. Artificial Radioactivity

Definition

Artificial radioactivity, or **induced radioactivity**, is the radioactivity produced artificially by bombarding a stable nucleus with particles such as neutrons or high-energy particles.

Simple idea

Stable nucleus → **bombardment** → **radioactive isotope** → **radioactive decay**

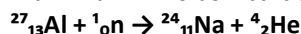
The important difference is:

Natural radioactivity: The unstable nucleus decays naturally.

Artificial radioactivity: Humans first **make a stable nucleus radioactive**.

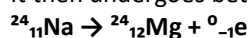
Example: Formation of Sodium-24

Aluminium-27 is bombarded with a neutron:



The sodium-24 produced is radioactive.

It then undergoes beta decay:



Thus, **Na-24** is an **artificially produced radioactive isotope**.

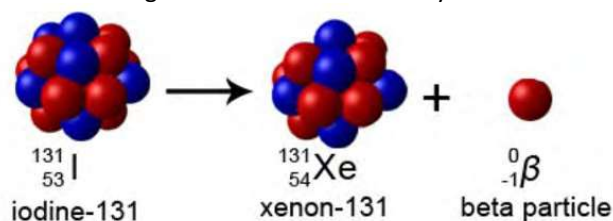
Examples of Artificial Radioisotopes

Stable element	Bombarded with	Radioactive isotope
Aluminium-27	Neutron	Sodium-24
Nitrogen-14	Alpha particle	Oxygen-17
Phosphorus-31	Neutron	Sulphur-32
Iodine-127	Neutron	Iodine-131

Applications of Artificial Radioactivity

1. Medicine

- **Iodine-131:** Diagnosis and treatment of thyroid disorders.



- **Cobalt-60:** Used in radiotherapy to destroy cancer cells.

2. Nuclear power

- **Plutonium-239:** Used as fuel in nuclear reactors.

3. Agriculture

- Used for studying plant growth and fertilizer uptake.
- Used in pest-control research.

4. Industry

- Used for **thickness measurement, leak detection and tracing**.

5. Scientific research

- Radioisotopes are used as **tracers** to study chemical, biological and physical processes.

Natural vs Artificial Radioactivity

Natural Radioactivity	Artificial Radioactivity
Occurs naturally	Produced artificially

No external trigger is required	Requires bombardment of a nucleus
Occurs in naturally radioactive isotopes	Produces radioactive isotopes artificially
Example: U-238	Example: Na-24, I-131
Used in dating, imaging and tracing	Widely used in medicine, industry, agriculture and research

Nuclear Reaction

Basic idea

A **nuclear reaction** is a process in which the **nucleus of an atom changes** and may produce a different nucleus/element, along with the release or absorption of a large amount of **energy and radiation**.

- **Chemical reaction:** involves mainly the **electrons outside the nucleus**.
- **Nuclear reaction:** involves the **nucleus itself**.
- Nuclear reactions generally release **much more energy** than chemical reactions.

Example: Energy of the Sun

The Sun produces its enormous energy through **nuclear fusion**, where light nuclei combine to form a heavier nucleus.

Deuterium + Tritium → Helium + Neutron + Energy

Nuclear equation:



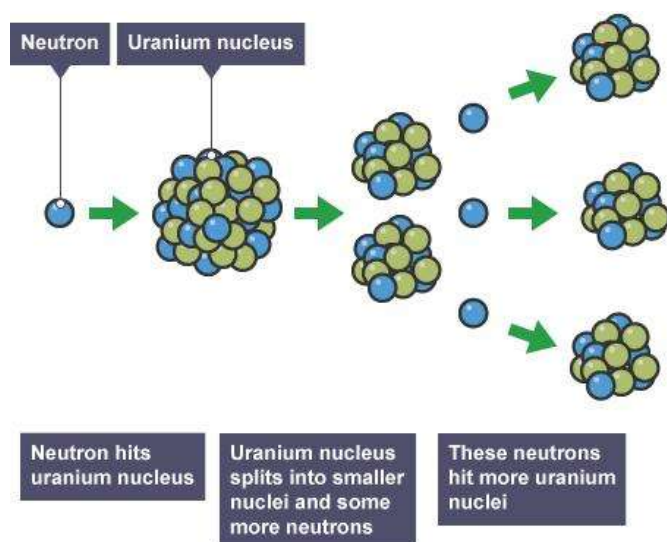
Here:

- ${}^2_1\text{H}$ = Deuterium, an isotope of hydrogen
- ${}^3_1\text{H}$ = Tritium, an isotope of hydrogen
- ${}^4_2\text{He}$ = Helium
- ${}^1_0\text{n}$ = Neutron

Types of Nuclear Reaction

There are **two main types**:

1. **Nuclear Fission**
2. **Nuclear Fusion**



1. Nuclear Fission

Meaning

Nuclear fission is a nuclear reaction in which a **heavy atomic nucleus splits into two or more lighter nuclei**, releasing a large amount of **energy and neutrons**.

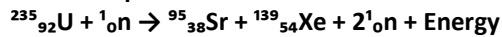
Think of it like:

Heavy nucleus → Smaller nuclei + neutrons + huge energy

Example: Fission of Uranium-235

When **Uranium-235** absorbs a neutron, it becomes unstable and splits into smaller nuclei.

Nuclear equation:



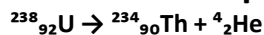
So:

Uranium-235 + neutron $\rightarrow$ Strontium-95 + Xenon-139 + 2 neutrons + energy

Important points

- A **heavy nucleus splits**.
- Produces **lighter nuclei**.
- Releases **large amount of energy**.
- Releases **neutrons**.
- The released neutrons can cause further fission, producing a **chain reaction**.
- Fission is used in **nuclear power plants** and **atomic weapons**.

Another example



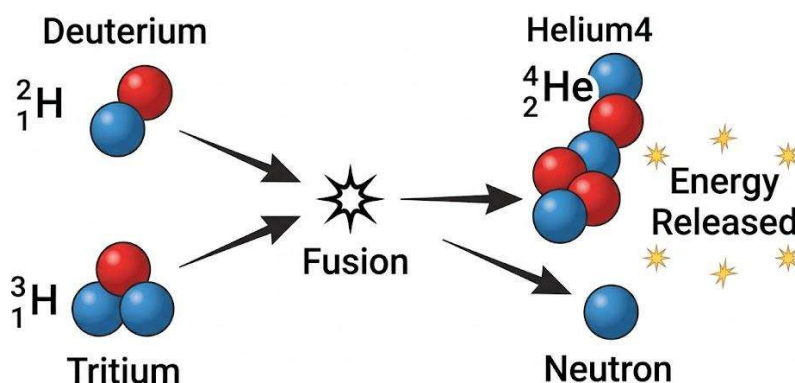
2. Nuclear Fusion

Meaning

Nuclear fusion is a nuclear reaction in which **two or more light atomic nuclei combine to form a heavier nucleus**, releasing a huge amount of energy.

Think of it as:

Small nuclei + Small nuclei $\rightarrow$ Larger nucleus + huge energy



Example: Deuterium + Tritium



Here, two hydrogen isotopes combine to produce **helium** and a **neutron**.

Important points

- **Light nuclei combine**.
- Produces a **heavier nucleus**.
- Releases a **very large amount of energy**.
- Occurs naturally in the **Sun and stars**.
- Does **not normally proceed as a chain reaction** like fission.
- Fusion has the potential to provide a **clean and almost limitless source of energy**.

Fission vs Fusion

Fission

Heavy nucleus **splits**

Produces smaller nuclei

Fusion

Light nuclei **combine**

Produces a heavier nucleus

Releases large energy	Releases even greater energy per reaction
Can produce a chain reaction	Does not normally produce a chain reaction
Used in nuclear power plants	Occurs naturally in the Sun and stars
Example: Uranium-235	Example: Deuterium + Tritium

Applications of Nuclear Fission

Nuclear fission is used for:

- **Generation of electricity** in nuclear power plants.
- Production of **radioisotopes** used in medicine.
- **Atomic weapons**.
- **Naval propulsion**, such as nuclear-powered submarines and aircraft carriers.

Naval propulsion

Heat produced by nuclear fission is used to produce **steam**. The steam drives turbines, which provide the power needed to move the vessel.

Applications of Nuclear Fusion

Nuclear fusion:

- Is the **source of energy of the Sun and stars**.
- Has potential to provide a **clean and virtually limitless source of energy**.
- Is being researched for **safe and sustainable electricity generation**.

Controlled and Uncontrolled Nuclear Fission

Controlled fission

A nuclear fission chain reaction in which the **rate of reaction is carefully controlled** so that energy is released at a steady and manageable rate.

Example: Nuclear power plant.

Uncontrolled fission

A chain reaction in which fission occurs **very rapidly without proper control**, releasing a huge amount of energy in a very short time.

Example: Atomic bomb.



Fig: Nuclear power station



Fig: Nuclear weapon

Nuclear Power and Nuclear Weapons

1. Nuclear Power

Nuclear power is the energy produced from **nuclear reactions**, mainly **nuclear fission**.

In nuclear fission, a heavy nucleus such as **Uranium-235** splits into smaller nuclei and releases a large amount of **heat energy**.

How nuclear power produces electricity

Nuclear fuel → Fission → Heat → Steam → Turbine → Generator → Electricity

- **Uranium-235** and **Plutonium-239** are commonly used as nuclear fuels.
- The heat produced during fission converts water into steam.
- Steam rotates turbines.

- Turbines drive generators to produce electricity.

Nuclear power is considered an important **low-carbon source of energy** because it produces very little carbon dioxide during electricity generation.

2. Nuclear Weapons

Nuclear weapons are explosive weapons that release an enormous amount of energy through **nuclear fission, fusion, or both**.

Types

Fission-based weapon

- Uses the splitting of heavy nuclei.
- Commonly called an **atomic bomb**.

Fusion-based weapon

- Uses the fusion of light nuclei.
- Also called a **hydrogen bomb or thermonuclear bomb**.
- Releases an extremely large amount of energy.

Nuclear weapons used in World War II

Two nuclear bombs were used against Japan in **1945**:

- **Little Boy** → dropped on **Hiroshima**
- **Fat Man** → dropped on **Nagasaki**

Both were **fission-based nuclear weapons**.

Nuclear Power vs Nuclear Weapons

Nuclear Power

Used for **electricity generation**

Nuclear reaction is **controlled**

Energy is released gradually

Useful for peaceful purposes

Nuclear Weapons

Used as **weapons of destruction**

Nuclear reaction is **uncontrolled/rapid**

Huge energy is released extremely rapidly

Causes massive destruction

Advantages of Nuclear Power

- Produces a **large amount of energy** from a small amount of fuel.
- Produces **very little carbon dioxide** during operation.
- Provides a reliable source of electricity.
- Requires relatively small amounts of fuel.
- Can provide continuous electricity for long periods.

Disadvantages of Nuclear Power

- Nuclear accidents can cause **long-term harm** to humans and the environment.
- Produces **radioactive nuclear waste** that requires careful long-term management.
- Nuclear power plants are **very expensive** to construct and maintain.
- Radiation exposure can be dangerous if safety systems fail.

Industrial Uses of Radioactivity

Radioisotopes and their radiation are useful in industries for **testing, measurement, leak detection, sterilization and process control**.

Commonly used radioisotopes include **Ir-192, Cs-137, Sr-90, Co-60 and Fe-55**.

1. Non-Destructive Testing (NDT)

Radioactive radiation is used to **detect internal defects** in materials without damaging them.

Example: Checking welds in aircraft parts, bridges and oil pipelines.

2. Thickness Gauging

Radiation is used to **measure and control the thickness** of materials during manufacturing.

Example: Beta radiation can be used to control the thickness of paper in paper mills.

3. Leak Detection

Radioactive substances can act as **tracers** to locate leaks in underground pipelines.

Basic idea:

Radioactive tracer → introduced into pipeline → detector detects radiation → leak location identified

4. Sterilization

Gamma radiation is used to kill microorganisms and sterilize certain products.

Example: Cobalt-60 is used for sterilizing medical equipment.

5. Power Generation in Remote Locations

Radioisotope Thermoelectric Generators (**RTGs**) convert heat from radioactive decay into electricity.

- Useful where ordinary power sources are difficult to use.
- Used in **space missions**.
- **Plutonium-238** is commonly used in RTGs.

6. Smoke Detectors

Some smoke detectors contain a small amount of **Americium-241 (Am-241)**.

- It produces ionizing radiation.
- Smoke changes the electrical conditions inside the detector.
- This triggers the alarm.

Medical Uses of Radioactivity

Radioisotopes are widely used in medicine for **diagnosis, imaging, treatment and biological research**.

1. Diagnostic Imaging: Lighting Up Organs from Within

- **SPECT & PET Scans:** Regular X-rays mainly show hard structures like bones. SPECT and PET scans allow doctors to see how organs are actively *functioning* (like tracking blood flow or sugar metabolism). When combined with a **CT scan**, they overlay this functional data onto a detailed 3D anatomical map.
- **Technetium-99m (Tc-99m): The Workhorse:** This is the most widely used medical isotope in the world. It emits clear gamma rays and has a very short half-life (about 6 hours), meaning it leaves the body quickly. Doctors attach it to specific chemicals that naturally travel to the **bones, heart, liver, or kidneys** to produce detailed images of organ health.
- **PET Scan Isotopes (Fluorine-18, Oxygen-15, Nitrogen-13):** These short-lived isotopes emit **positrons** (positive antimatter electrons). When injected, the positron travels a tiny distance before interacting with a regular negative electron inside the tissue. They annihilate each other, producing **two gamma rays in opposite directions**. The PET scanner detects these signals to determine the location of tumors or areas of brain activity.

2. Cancer Treatment: Blasting Bad Cells

- **Iodine-131 (I-131):** The thyroid gland naturally absorbs iodine to make hormones. When a patient takes I-131, it travels mainly to the thyroid. Its localized beta radiation can destroy overactive thyroid tissue or thyroid cancer cells.
- **Cobalt-60 (Co-60):** This isotope is used as an external radiation source. It produces high-energy gamma rays that can be directed at deep-seated tumors, damaging the DNA of cancer cells so they cannot continue dividing.
- **Bismuth-213 (Bi-213):** Used in **Targeted Alpha Therapy (TAT)**. It can be attached to molecules that target specific cancer-cell proteins. It then releases powerful alpha particles over a very short distance, damaging nearby cancer cells.

3. Brachytherapy: An Inside-Out Defense

- **How it works:** Instead of directing radiation from outside the body, brachytherapy involves placing tiny radioactive **seeds, wires, or needles directly inside or close to the tumor**.
- **The Benefit:** It is used for localized cancers such as **prostate, breast, cervical, and skin cancer**. A high radiation dose can be delivered directly to the tumor while reducing radiation exposure to surrounding healthy tissues.
- **Isotopes Used:** **Iridium-192 (Ir-192)** is commonly used in temporary implants, while

Radium-223 (Ra-223) can be used for certain cancers that have spread to bones.

4. Iron Metabolism: Tracking Your Blood Builder

- **Iron-59 (Fe-59):** Iron is essential for producing hemoglobin, which carries oxygen in the blood.
- **The Application:** A small amount of radioactive Iron-59 can be used as a tracer to study how the body absorbs, transports, and stores iron. It can help researchers study iron movement through organs such as the **spleen and liver** and its use by the **bone marrow** to produce red blood cells.

5. Blood Flow: Checking the Plumbing

- **Sodium-24 (Na-24):** Sodium is naturally present in blood and can be used as a radioactive tracer.
- **The Application:** Sodium-24 can be introduced into the bloodstream as a tracer. Its movement can then be monitored using radiation detectors to study **blood circulation and blood flow**.

6. Biological Research: The Invisible Lab Labels

- **Sulfur-35 (S-35) & Phosphorus-32 (P-32):** In molecular biology, radioactive isotopes can be used as **tracers** to follow molecules that cannot be directly observed.
- **Labeling Proteins & DNA:** Phosphorus is an important component of DNA and RNA, while sulfur is present in certain proteins. Radioactive P-32 or S-35 can therefore be used to label biological molecules.
- **Tracking Life Processes:** This allows researchers to study processes such as **DNA/RNA activity, protein synthesis, drug interactions, and metabolic pathways**, making radioactive tracers valuable tools in biological research.

Radiocarbon Dating

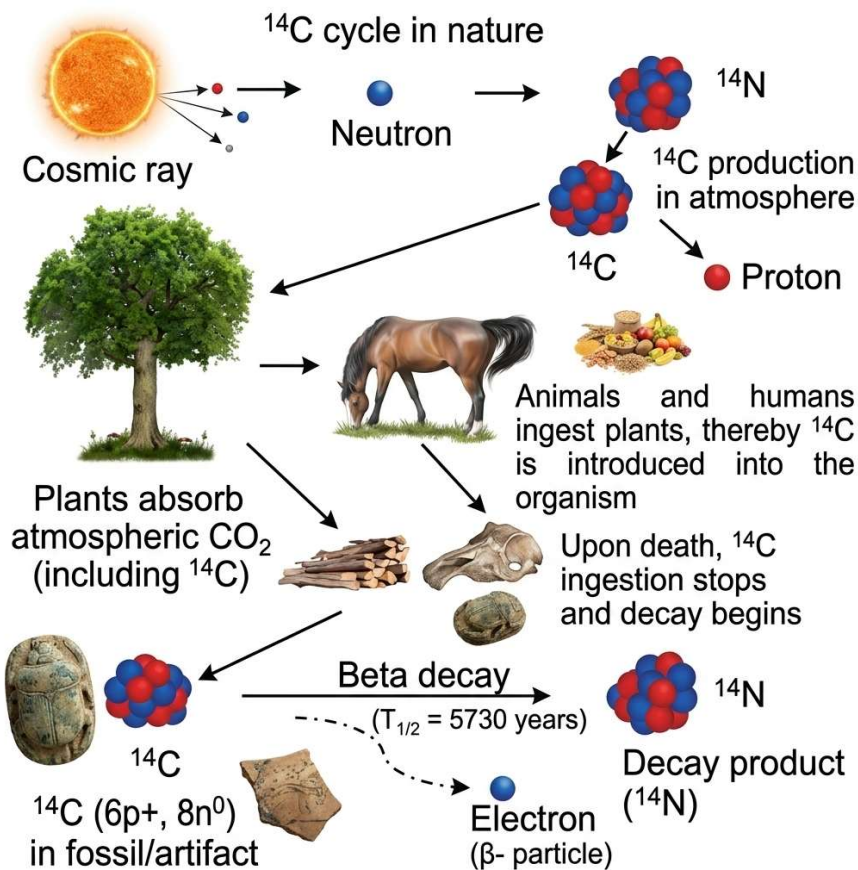
Meaning

Radiocarbon dating is a method used to determine the age of **once-living materials** by measuring the amount of radioactive **Carbon-14 (C-14)** remaining in them.

It is mainly used in **archaeology, geology and paleontology** to estimate the age of fossils, bones, wood, cloth and other ancient biological materials.

Discovery

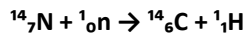
- **Willard Frank Libby** developed radiocarbon dating in **1949**.
- He received the **Nobel Prize in Chemistry in 1960** for this work.



How Radiocarbon Dating Works

1. Formation of Carbon-14

Cosmic rays produce neutrons in the atmosphere. These neutrons react with **nitrogen-14** to produce **Carbon-14**.



Carbon-14 is then converted into **CO₂** in the atmosphere.

2. Carbon-14 enters living organisms

- Plants absorb CO₂ during **photosynthesis**.
- Therefore, plants contain C-14.
- Animals obtain C-14 by eating plants or other animals.
- While an organism is alive, it continuously exchanges carbon with its environment, so its C-14 level remains approximately constant.

3. After death

When the organism dies:

No more C-14 enters → C-14 gradually decays → amount of C-14 decreases

The C-14 already present in the organism continues to undergo radioactive decay.

4. Determining the age

Scientists measure the **amount of C-14 remaining** in the dead material.

- **More C-14 remaining → younger sample**
- **Less C-14 remaining → older sample**

The half-life of C-14 is:

Half-life = 5730 years

This means that every **5730 years**, approximately half of the remaining C-14 decays.

Formula for Radiocarbon Dating

The age of the sample can be calculated using:

$$t = \frac{t_{1/2}}{\log 2} \log \left(\frac{N_0}{N} \right)$$

Where:

- **t** = age of the sample
- **t_{1/2}** = half-life of C-14 = **5730 years**
- **N₀** = original amount of C-14 when the organism was alive
- **N** = amount of C-14 remaining in the sample

Important point

C-14 undergoes **first-order radioactive decay**, meaning its decay rate depends on the amount of C-14 remaining.

Harmful Effects of Nuclear Radiations

Nuclear radiation has many useful applications, but excessive exposure can seriously damage living organisms.

Major harmful effects

1. Cell and DNA damage
2. Increased risk of cancer
3. Radiation burns and radiation sickness
4. Infertility and birth defects
5. Weakening of the immune system
6. Environmental contamination